Dual variables \ Primal variables	$x_1 \geq 0$	$x_2 \geq 0$	$x_3 \geq 0$	$\cdots$	$x_n \geq 0$	Primal relation	Min v
$y_1 \geqq 0$	a_{11}	a_{12}	a_{13}	$\cdots$	a_{1n}	$\leq$	b_1
$y_2 \geqq 0$	a_{21}	a_{22}	a_{23}	$\cdots$	a_{2n}	$\leq$	b_2
$\vdots$	$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$	$\vdots$
$y_m \geqq 0$	a_{m1}	a_{m2}	a_{m3}	$\cdots$	a_{mn}	$\leq$	b_m
Dual Relation	$\geq$	$\geq$	$\geq$		$\geq$		
Max z	c_1	c_2	c_3	$\cdots$	c_n		

Primal (Maximize)	*Dual (Minimize)*
ith constraint $\leq$	ith variable ≥ 0
ith constraint $\geq$	ith variable ≤ 0
ith constraint $=$	ith variable unrestricted
jth variable ≥ 0	jth constraint $\geq$
jth variable ≤ 0	jth constraint $\leq$
jth variable unrestricted	jth constraint $=$

Table 4.2

1	*Primal feasible* Maximize $z = 2x_1 + x_2$, subject to: $x_1 + x_2 \leq 4$, $x_1 - x_2 \leq 2$, $x_1 \geq 0, \quad x_2 \geq 0.$	*Dual feasible* Minimize $v = 4y_1 + 2y_2$, subject to: $y_1 + y_2 \geq 2$, $y_1 - y_2 \geq 1$, $y_1 \geq 0, \quad y_2 \geq 0.$
2	*Primal feasible and unbounded* Maximize $z = 2x_1 + x_2$, subject to: $x_1 - x_2 \leq 4$, $x_1 - x_2 \leq 2$, $x_1 \geq 0, \quad x_2 \geq 0.$	*Dual infeasible* Minimize $v = 4y_1 + 2y_2$, subject to: $y_1 + y_2 \geq 2$, $-y_1 - y_2 \geq 1$, $y_1 \geq 0, \quad y_2 \geq 0.$
	Primal infeasible Maximize $z = 2x_1 + x_2$, subject to: $-x_1 - x_2 \leq -4$, $x_1 + x_2 \leq 2$, $x_1 \geq 0, \quad x_2 \geq 0.$	*Dual feasible and unbounded* Minimize $v = -4y_1 + 2y_2$, subject to: $-y_1 + y_2 \geq 2$, $-y_1 + y_2 \geq 1$, $y_1 \geq 0, \quad y_2 \geq 0.$
	Primal infeasible Maximize $z = 2x_1 + x_2$, subject to: $-x_1 + x_2 \leq -4$, $x_1 - x_2 \leq 2$, $x_1 \geq 0, \quad x_2 \geq 0.$	*Dual infeasible* Minimize $v = -4y_1 + 2y_2$, subject to: $-y_1 + y_2 \geq 2$, $y_1 - y_2 \geq 1$, $y_1 \geq 0, \quad y_2 \geq 0.$

BV_i = basic variable for ith constraint in the optimal tableau

$\mathbf{c}_{\text{BV}}$ = row vector whose ith element is the objective function c the LP

$\mathbf{a}_j$ = column for variable x_j in constraints of original LP

$\mathbf{b}$ = right-hand side vector for original LP

$\bar{c}_j$ = coefficient of x_j in row 0 of the optimal tableau

Duality and Sensitivity Analysis

Our proof of the Dual Theorem showed that if a set of basic variables BV is feasible, then BV is optimal (that is, each variable in row 0 has a nonnegative coefficient) if and only if the associated dual solution, $\mathbf{c}_{\text{BV}}B^{-1}$, is dual feasible.

This result can be used to yield an alternative way of doing the following types of sensitivity analysis:

Change 1 Changing the objective function coefficient of a nonbasic variable

Change 4 Changing the column of a nonbasic variable

Change 5 Adding a new activity

How to Compute Optimal Tableau from Initial LP

Column for x_j in optimal tableau's constraints $= B^{-1}\mathbf{a}_j$ **(5)**

Right-hand side of optimal tableau's constraints $= B^{-1}\mathbf{b}$ **(6)**

$\bar{c}_j = \mathbf{c}_{\text{BV}}B^{-1}\mathbf{a}_j - c_j$ **(10)**

Coefficient of slack variable s_i in optimal row 0 = ith element of $\mathbf{c}_{\text{BV}}B^{-1}$ **(10′)**

Coefficient of excess variable e_i in optimal row 0 = $-(i\text{th element of } \mathbf{c}_{\text{BV}}B^{-1})$ **(10″)**

Coefficient of artificial variable a_i in optimal row 0 = $(i\text{th element of } \mathbf{c}_{\text{BV}}B^{-1}) + M$ **(10‴)**

Right-hand side of optimal row 0 $= \mathbf{c}_{\text{BV}}B^{-1}\mathbf{b}$ **(11)**

Shadow Prices (Again)

For a maximization LP, the shadow price of the ith constraint is the value of the ith dual variable in the optimal dual solution. For a minimization LP, the shadow price of the ith constraint = $-$(ith dual variable in the optimal dual solution). The shadow price of the ith constraint is found in row $i + 1$ of the DUAL PRICES portion of the LINDO printout.

New optimal z-value = (old optimal z-value) + (Constraint i shadow price) Δb_i (max problem) **(37)**

New optimal z-value = (old optimal z-value) $-$ (Constraint i shadow price) Δb_i (min problem) **(37′)**

A $\geq$ constraint will have a nonpositive shadow price; a $\leq$ constraint will have a nonnegative shadow price; and an equality constraint may have a positive, negative, or zero shadow price.

Finding the Optimal Solution to the Dual of an LP

If the primal is a max problem, then the optimal dual solution may be read from row 0 of the optimal tableau by using the following rules:

Optimal value of dual variable y_i if Constraint i is a $\leq$ constraint = coefficient of s_i in optimal row 0 **(31)**

Optimal value of dual variable y_i if Constraint i is a $\geq$ constraint = $-$(coefficient of e_i in optimal row 0) **(31′)**

Optimal value of dual variable y_i if Constraint i is an equality constraint = (coefficient of a_i in optimal row 0) $- M$ **(31″)**

If the primal is a min problem, then the optimal dual solution may be read from row 0 of the optimal tableau by using the following rules:

Optimal value of dual variable x_i if Constraint i is a $\leq$ constraint = coefficient of s_i in optimal row 0

Optimal value of dual variable x_i if Constraint i is a $\geq$ constraint = $-$(coefficient of e_i in optimal row 0)

Optimal value of dual variable x_i if Constraint i is an equality constraint = (coefficient of a_i in optimal row 0) $+ M$

TABLE 9

Summary of Sensitivity Analysis (Max Problem)

Change in Initial Problem	Effect on Optimal Tableau	Current Basis Is Still Optimal If:
Changing nonbasic objective function coefficient c_j	Coefficient of x_j in optimal row 0 is changed	Coefficient of x_j in row 0 for current basis is still nonnegative
Changing basic objective function coefficient c_j	Entire row 0 may change	Each variable still has a nonnegative coefficient in row 0
Changing right-hand side of a constraint	Right-hand side of constraints and row 0 are changed	Right-hand side of each constraint is still nonnegative
Changing the column of a nonbasic variable x_j or adding a new variable x_j	Changes the coefficient for x_j in row 0 and x_j's constraint column in optimal tableau	The coefficient of x_j in row 0 is still nonnegative

The Dual Simplex Method

The dual simplex method can be applied (to a max problem) whenever there is a basic solution in which each variable has a nonnegative coefficient in row 0. If we have found such a basic solution, then the dual simplex method proceeds as follows:

Step 1 If the right-hand side of each constraint is nonnegative, then an optimal solution has been found; if not, then at least one constraint has a negative right-hand side, and we go to step 2.

Step 2 Choose the most negative basic variable as the variable to leave the basis. The row in which this variable is basic will be the pivot row. To select the variable that enters the basis, compute the following ratio for each variable x_j that has a *negative* coefficient in the pivot row:

$$\frac{\text{Coefficient of } x_j \text{ in row 0}}{}$$

Choose the variable that has the smallest ratio (absolute value) as the entering variable. Use EROs to make the entering variable a basic variation in the pivot row.

Step 3 If there is any constraint in which the right-hand side is negative and each variable has a nonnegative coefficient, then the LP has no feasible solution. Infeasibility would be indicated by the presence (after possibly several pivots) of a constraint such as $x_1 + 2x_2 + x_3 = -5$. If no constraint indicating infeasibility is found, return to step 1.

The dual simplex method is often used in the following situations:

1. Finding the new optimal solution after a constraint is added to an LP
2. Finding the new optimal solution after changing an LP's right-hand side
3. Solving a normal min problem

A Diet Problem

$$\text{Cost of cola} = \left(\frac{\text{cost}}{\text{bottle of cola}}\right)\left(\begin{array}{c}\text{bottles of}\\ \text{cola drunk}\end{array}\right) = 30x_3$$

Type of Food	Calories	Chocolate (Ounces)	Sugar (Ounces)	Fat (Ounces)
Brownie	400	3	2	2
Chocolate ice cream (1 scoop)	200	2	2	4
Cola (1 bottle)	150	0	4	1
Pineapple cheesecake (1 piece)	500	0	4	5

Constraint 1 Daily calorie intake must be at least 500 calories.

Constraint 2 Daily chocolate intake must be at least 6 oz.

Constraint 3 Daily sugar intake must be at least 10 oz.

Constraint 4 Daily fat intake must be at least 8 oz.

$$\min z = 50x_1 + 20x_2 + 30x_3 + 80x_4$$

$$\begin{array}{llr} \text{s.t.} \quad 400x_1 + 200x_2 + 150x_3 + 500x_4 \ge 500 & \text{(Calorie constraint)} & (21)\\ 3x_1 + 2x_2 \ge 6 & \text{(Chocolate constraint)} & (22)\\ 2x_1 + 2x_2 + 4x_3 + 4x_4 \ge 10 & \text{(Sugar constraint)} & (23)\\ 2x_1 + 4x_2 + x_3 + 5x_4 \ge 8 & \text{(Fat constraint)} & (24)\\ x_i \ge 0 \ (i = 1, 2, 3, 4) & \text{(Sign restrictions)} & \end{array}$$

Case 1 The LP has a unique optimal solution.

Case 2 The LP has alternative or multiple optimal solutions: Two or more extreme points are optimal, and the LP will have an infinite number of optimal solutions.

Case 3 The LP is infeasible: The feasible region contains no points.

Case 4 The LP is unbounded: There are points in the feasible region with arbitrarily large z-values (max problem) or arbitrarily small z-values (min problem).

5 Let A = total number of units of A produced

B = total number of units of B produced

CS = total number of units of C produced (and sold)

AS = units of A sold

BS = units of B sold

$$\max z = 10AS + 56BS + 100CS$$

$$\begin{aligned} \text{s.t.} \quad A + 2B + 3C &\le 40\\ A &= AS + 2B\\ B &= BS + CS \end{aligned}$$

All variables ≥ 0

A Work-Scheduling Problem

1 Let x_1 = number of full-time employees (FTE) who start work on Sunday, x_2 = number of FTE who start work on Monday, . . . , x_7 = number of FTE who start work on Saturday; x_8 = number of part-time employees (PTE) who start work on Sunday, . . . , x_{14} = number of PTE who start work on Saturday. Then the appropriate LP is

$$\min z = 15(8)(5)(x_1 + x_2 + \cdots + x_7) + 10(4)(5)(x_8 + x_9 + \cdots + x_{14})$$

$$\begin{array}{lr} \text{s.t.} \quad 8(x_1 + x_4 + x_5 + x_6 + x_7) + 4(x_8 + x_{11} + x_{12} + x_{13} + x_{14}) \ge 88 & \text{(Sunday)}\\ 8(x_1 + x_2 + x_5 + x_6 + x_7) + 4(x_8 + x_9 + x_{12} + x_{13} + x_{14}) \ge 136 & \text{(Monday)}\\ 8(x_1 + x_2 + x_3 + x_6 + x_7) + 4(x_8 + x_9 + x_{10} + x_{13} + x_{14}) \ge 104 & \text{(Tuesday)}\\ 8(x_1 + x_2 + x_3 + x_4 + x_7) + 4(x_8 + x_9 + x_{10} + x_{11} + x_{14}) \ge 120 & \text{(Wednesday)}\\ 8(x_1 + x_2 + x_3 + x_4 + x_5) + 4(x_8 + x_9 + x_{10} + x_{11} + x_{12}) \ge 152 & \text{(Thursday)}\\ 8(x_2 + x_3 + x_4 + x_5 + x_6) + 4(x_9 + x_{10} + x_{11} + x_{12} + x_{13}) \ge 112 & \text{(Friday)}\\ 8(x_3 + x_4 + x_5 + x_6 + x_7) + 4(x_{10} + x_{11} + x_{12} + x_{13} + x_{14}) \ge 128 & \text{(Saturday)} \end{array}$$

$$20(x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14}) \le 0.25(136 + 104 + 120 + 152 + 112 + 128 + 88)$$

(The last constraint ensures that part-time labor will fulfill at most 25% of all labor requirements)

All variables ≥ 0

Production Process Models

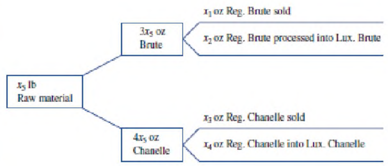
1 You have decided to enter the candy business. You are considering producing two types of candies: Slugger Candy and Easy Out Candy, both of which consist solely of sugar, nuts, and chocolate. At present, you have in stock 100 oz of sugar, 20 oz of nuts, and 30 oz of chocolate. The mixture used to make Easy Out Candy must contain at least 20% nuts. The mixture used to make Slugger Candy must contain at least 10% nuts and 10% chocolate. Each ounce of Easy Out Candy can be sold for 25¢ , and each ounce of Slugger Candy for 20¢. Formulate an LP that will enable you to maximize your revenue from candy sales.

1 Let ingredient 1 = sugar, ingredient 2 = nuts, ingredient 3 = chocolate, candy 1 = Slugger, and candy 2 = Easy Out. Let x_{ij} = ounces of ingredient i used to make candy j. (All variables are in ounces.) The appropriate LP is

$$\max z = 25(x_{12} + x_{22} + x_{32}) + 20(x_{11} + x_{21} + x_{31})$$

$$\begin{array}{ll} \text{s.t.} \quad x_{11} + x_{12} \le 100 & \text{(Sugar constraint)}\\ x_{21} + x_{22} \le 20 & \text{(Nuts constraint)}\\ x_{31} + x_{32} \le 30 & \text{(Chocolate constraint)}\\ x_{22} \ge 0.2(x_{12} + x_{22} + x_{32}) & \\ x_{21} \le 0.1(x_{11} + x_{21} + x_{31}) & \\ x_{31} \ge 0.1(x_{11} + x_{21} + x_{31}) & \end{array}$$

All variables ≥ 0



Blending Problem

1 Let x_1 = hours of process 1 run per week

x_2 = hours of process 2 run per week

x_3 = hours of process 3 run per week

g_2 = barrels of gas 2 sold per week

o_1 = barrels of oil 1 purchased per week

o_2 = barrels of oil 2 purchased per week

$$\max z = 9(2x_1) + 10g_2 + 24(2x_3) - 5x_1 - 4x_2 - x_3 - 2o_1 - 3o_2$$

$$\begin{array}{ll} \text{s.t.} \quad o_1 = 2x_1 + x_2 & \\ o_2 = 3x_1 + 3x_2 + 2x_3 & \\ o_1 \le 200 & \\ o_2 \le 300 & \\ g_2 + 3x_3 = x_1 + 3x_2 & \text{(Gas 2 production)}\\ x_1 + x_2 + x_3 \le 100 & \text{(100 hours per week of cracker time)} \end{array}$$

All variables ≥ 0

x_1 = number of ounces of Regular Brute sold annually

x_2 = number of ounces of Luxury Brute sold annually

x_3 = number of ounces of Regular Chanelle sold annually

x_4 = number of ounces of Luxury Chanelle sold annually

x_5 = number of pounds of raw material purchased annually

$$\max z = 7x_1 + 14x_2 + 6x_3 + 10x_4 - 3x_5$$

$$\begin{aligned} \text{s.t.} \quad x_5 &\le 4{,}000\\ 3x_2 + 2x_4 + x_5 &\le 6{,}000\\ x_1 + x_2 - 3x_5 &= 0\\ x_3 + x_4 - 4x_5 &= 0\\ x_i \ge 0 \quad (i &= 1, 2, 3, 4, 5) \end{aligned}$$

An Inventory Model

1 A customer requires during the next four months, respectively, 50, 65, 100, and 70 units of a commodity (no backlogging is allowed). Production costs are \$5, \$8, \$4, and \$7 per unit during these months. The storage cost from one month to the next is \$2 per unit (assessed on ending inventory). It is estimated that each unit on hand at the end of month 4 could be sold for \$6. Formulate an LP that will minimize the net cost incurred in meeting the demands of the next four months.

Let x_t = production during month t and i_t = inver t end of month t.

$$\min z = 5x_1 + 8x_2 + 4x_3 + 7x_4 + 2i_1 + 2i_2 + 2i_3 + 2i_4 - 6i_4$$

$$\begin{aligned} \text{s.t.} \quad i_1 &= x_1 - 50\\ i_2 &= i_1 + x_2 - 65\\ i_3 &= i_2 + x_3 - 100\\ i_4 &= i_3 + x_4 - 70 \end{aligned}$$

All variables ≥ 0

$$\boldsymbol{i_t - b_t = i_{t-1} - b_{t-1} + X_t - d_t}$$

3 Let x_1 = number of employees who start work on Sunday and work five days, x_2 = number of employees who start work on Monday and work five days, . . . , x_7 = number of employees who start work on Saturday and work five days. Also let o_1 = number of employees who start work on Sunday and work six days, . . . , o_7 = number of employees who start work on Saturday and work six days. Then the appropriate LP is

$$\min z = 250(x_1 + x_2 + \cdots + x_7) + 312(o_1 + o_2 + \cdots + o_7)$$

$$\begin{array}{lr} \text{s.t.} \quad x_1 + x_4 + x_5 + x_6 + x_7 + o_1 + o_3 + o_4 + o_5 + o_6 + o_7 \ge 11 & \text{(Sunday)}\\ x_1 + x_2 + x_5 + x_6 + x_7 + o_1 + o_2 + o_4 + o_5 + o_6 + o_7 \ge 17 & \text{(Monday)}\\ x_1 + x_2 + x_3 + x_6 + x_7 + o_1 + o_2 + o_3 + o_5 + o_6 + o_7 \ge 13 & \text{(Tuesday)}\\ x_1 + x_2 + x_3 + x_4 + x_7 + o_1 + o_2 + o_3 + o_4 + o_6 + o_7 \ge 15 & \text{(Wednesday)}\\ x_1 + x_2 + x_3 + x_4 + x_5 + o_1 + o_2 + o_3 + o_4 + o_5 + o_7 \ge 19 & \text{(Thursday)}\\ x_2 + x_3 + x_4 + x_5 + x_6 + o_1 + o_2 + o_3 + o_4 + o_5 + o_6 \ge 14 & \text{(Friday)}\\ x_3 + x_4 + x_5 + x_6 + x_7 + o_2 + o_3 + o_4 + o_5 + o_6 + o_7 \ge 16 & \text{(Saturday)} \end{array}$$

All variables ≥ 0

Solution:

<u>Define variables:</u>

C1 = number of cars to be produced in month 1

C2 = number of cars to be produced in month 2

T1 = number of trucks to be produced in month 1

T2 = number of truckes to be produced in month 2

S1 = tons of steel used in month 1

S2 = tons of steel used in month 2

IC1 = number of cars in inventory at end of month 1

IT1 = number of trucks in inventory at end of month 1

IC2 = number of cars in inventory at end of month 2

IT2 = number of trucks in inventory at end of month 2

<u>Objective:</u>

Minimize 400 S1 + 600 S2 + 150 IC1 + 150 IT1 + 150 IC2 + 150 IT2

<u>Constraints:</u>

Production capacity constraints: C1 + T1 £ 1000, C2 + T2 £ 1000

Steel usage: C1 + 2 T1 = S1, C2 + 2 T2 = S2

Material balance equations: 200 + C1 = 800 + IC1, 100 + T1 = 400 + IT1

IC1 + C2 = 300 + IC2, IT1 + T2 = 300 + IT2

Demand constraints: C2 ³ 300, T2 ³ 300

Gasoline economy constraints:

$\dfrac{20C1 + 10T1}{C1 + T1}$ ³ 16 Þ 4C1 − 6T1 ³ 0

$\dfrac{20C2 + 10T2}{C2 + T2}$ ³ 16 Þ 4C2 − 6T2 ³ 0

	Cash Flow (\$) at Time*			
	0	1	2	3
A	−1	+0.50	+1	0
B	0	−1	+0.50	+1
C	−1	+1.2	0	0
D	−1	0	0	+1.9
E	0	0	−1	+1.5

*Note: Time 0 = present; time 1 = 1 year from now; time 2 = 2 years from now; time 3 = 3 years from now.

53 Let T_1 = number of type 1 turkeys purchased

T_2 = number of type 2 turkeys purchased

D_1 = pounds of dark meat used in cutlet 1

W_1 = pounds of white meat used in cutlet 1

D_2 = pounds of dark meat used in cutlet 2

W_2 = pounds of white meat used in cutlet 2

Then the appropriate formulation is

$$\max z = 4(W_1 + D_1) + 3(W_2 + D_2) - 10T_1 - 8T_2$$

$$\begin{array}{ll} \text{s.t.} \quad W_1 + D_1 \le 50 & \text{(Cutlet 1 demand)}\\ W_2 + D_2 \le 30 & \text{(Cutlet 2 demand)}\\ W_1 + W_2 \le 5T_1 + 3T_2 & \text{(Don't use more white meat than you have)}\\ D_1 + D_2 \le 2T_1 + 3T_2 & \text{(Don't use more dark meat than you have)} \end{array}$$

A = dollars invested in investment A

B = dollars invested in investment B

C = dollars invested in investment C

D = dollars invested in investment D

E = dollars invested in investment E

S_t = dollars invested in money market

$$\max z = B + 1.9D + 1.5E + 1.08S_2$$

$$\begin{aligned} \text{s.t.} \quad A + C + D + S_0 &= 100{,}000\\ 0.5A + 1.2C + 1.08S_0 &= B + S_1\\ A + 0.5B + 1.08S_1 &= E + S_2\\ A &\le 75{,}000\\ B &\le 75{,}000\\ C &\le 75{,}000\\ D &\le 75{,}000\\ E &\le 75{,}000\\ A, B, C, D, E, S_0, S_1, S_2 &\ge 0 \end{aligned}$$